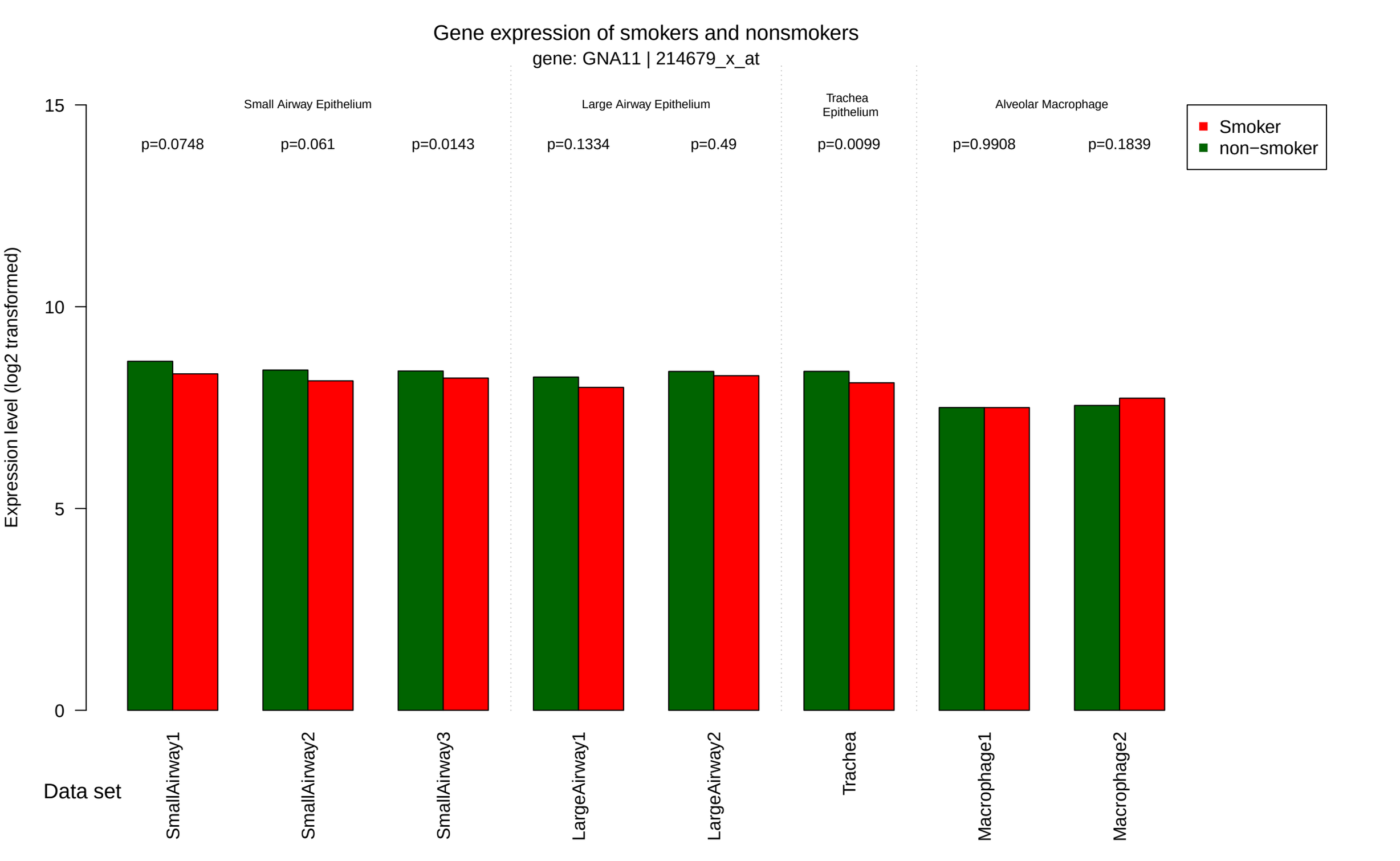
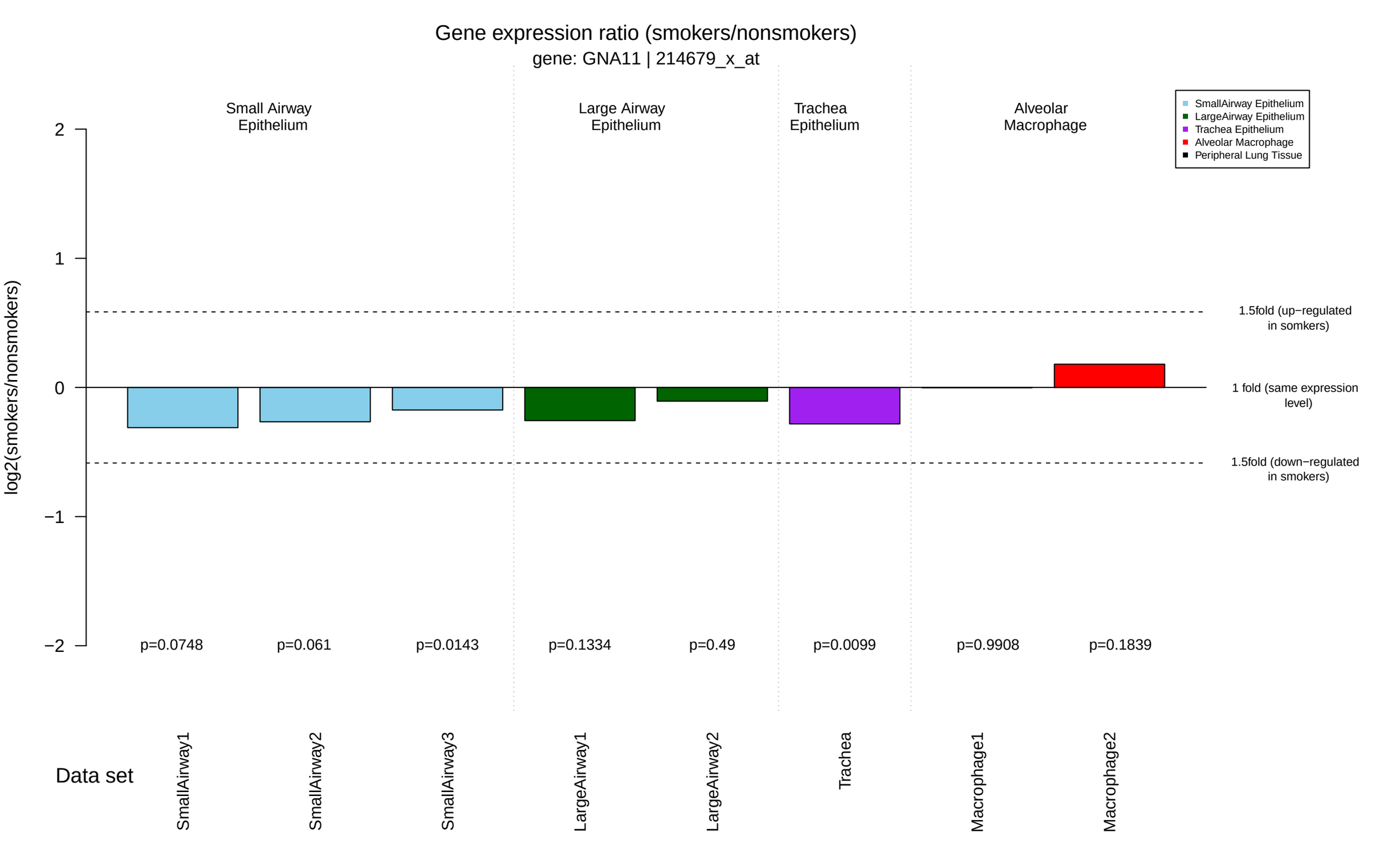
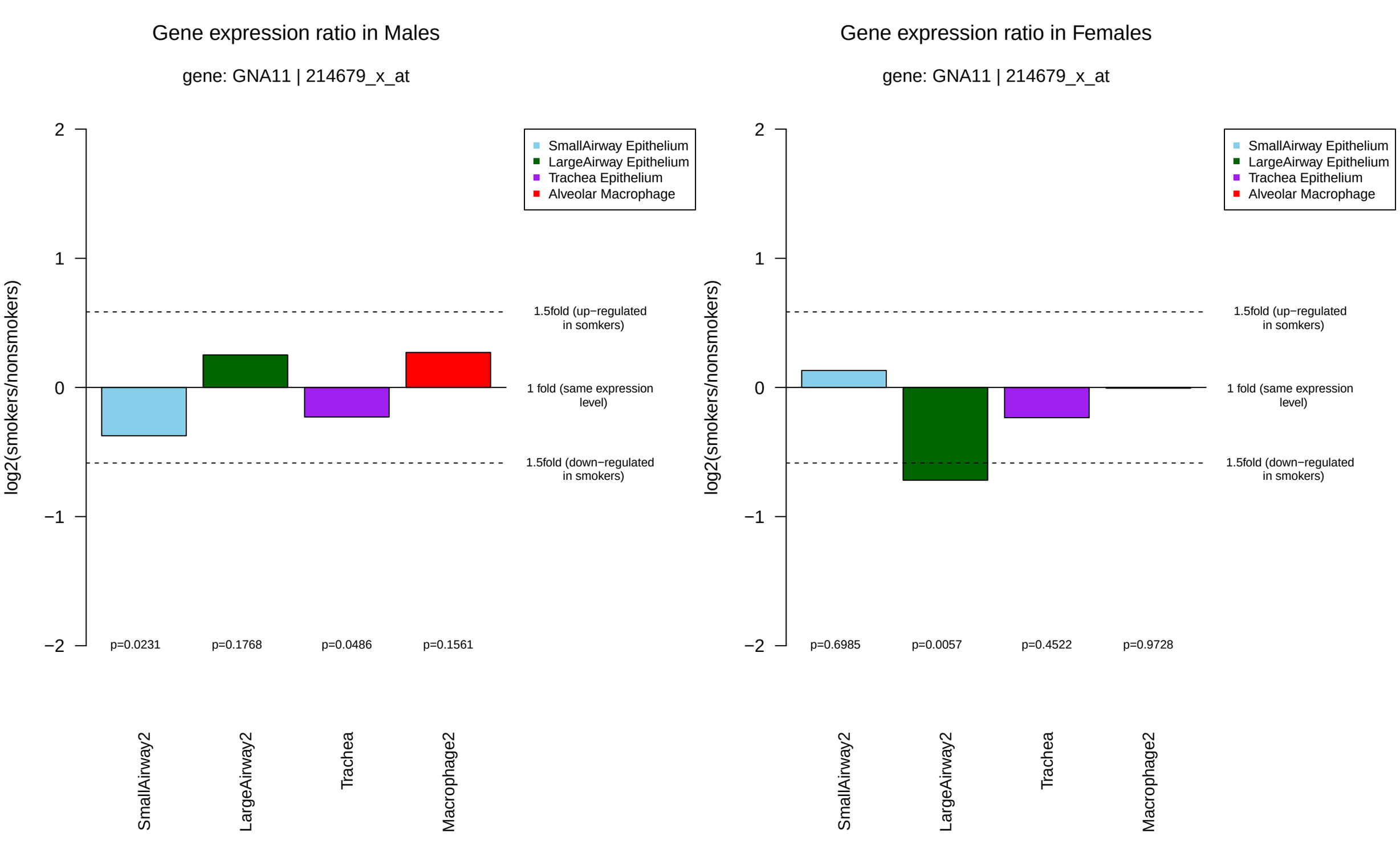
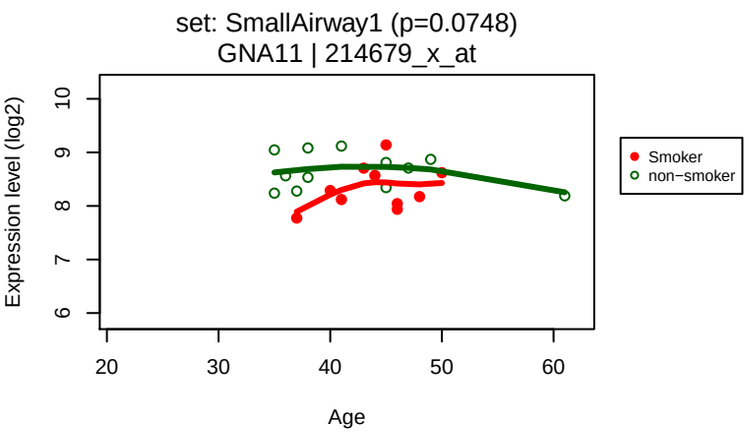
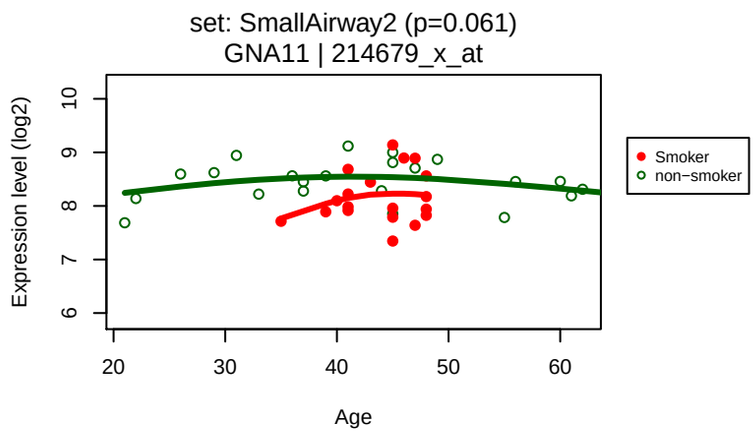
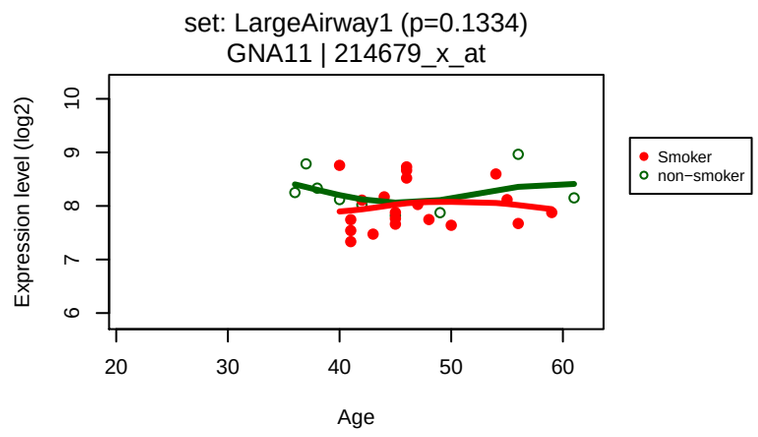

Gene expression of smokers and nonsmokers
gene: GNA11 | 214679_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0748
p=0.061
p=0.0143
p=0.1334
p=0.49
p=0.0099
p=0.9908
p=0.1839
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

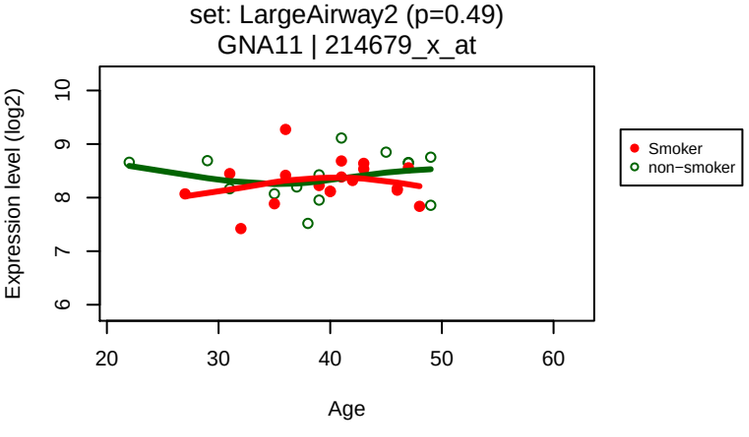

Gene expression ratio (smokers/nonsmokers)
gene: GNA11 | 214679_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0748
p=0.061
p=0.0143
p=0.1334
p=0.49
p=0.0099
p=0.9908
p=0.1839
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

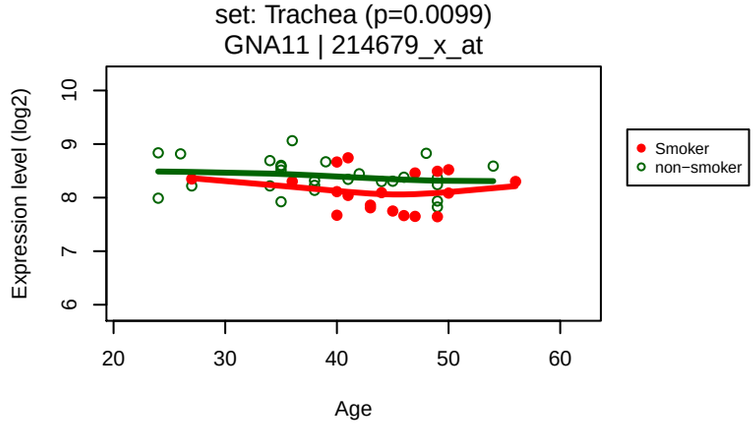

Gene expression ratio in Males
gene: GNA11 | 214679_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0231
p=0.1768
p=0.0486
p=0.1561
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNA11 | 214679_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6985
p=0.0057
p=0.4522
p=0.9728
SmallAirway2
LargeAirway2
Trachea
Macrophage2

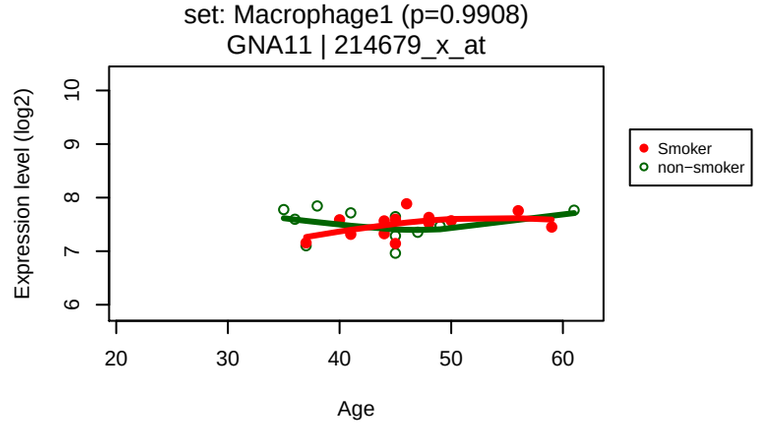

set: SmallAirway1 (p=0.0748)
GNA11 | 214679_x_at
Expression level (log2)
Age
Smoker
non−smoker

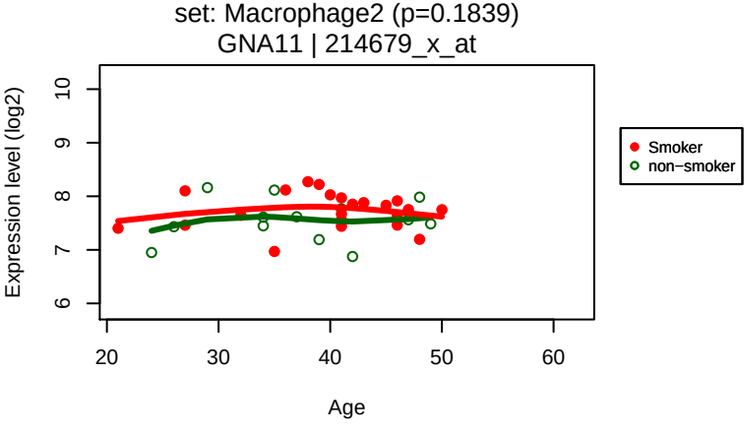

set: SmallAirway2 (p=0.061)
GNA11 | 214679_x_at
Expression level (log2)
Age
Smoker
non−smoker

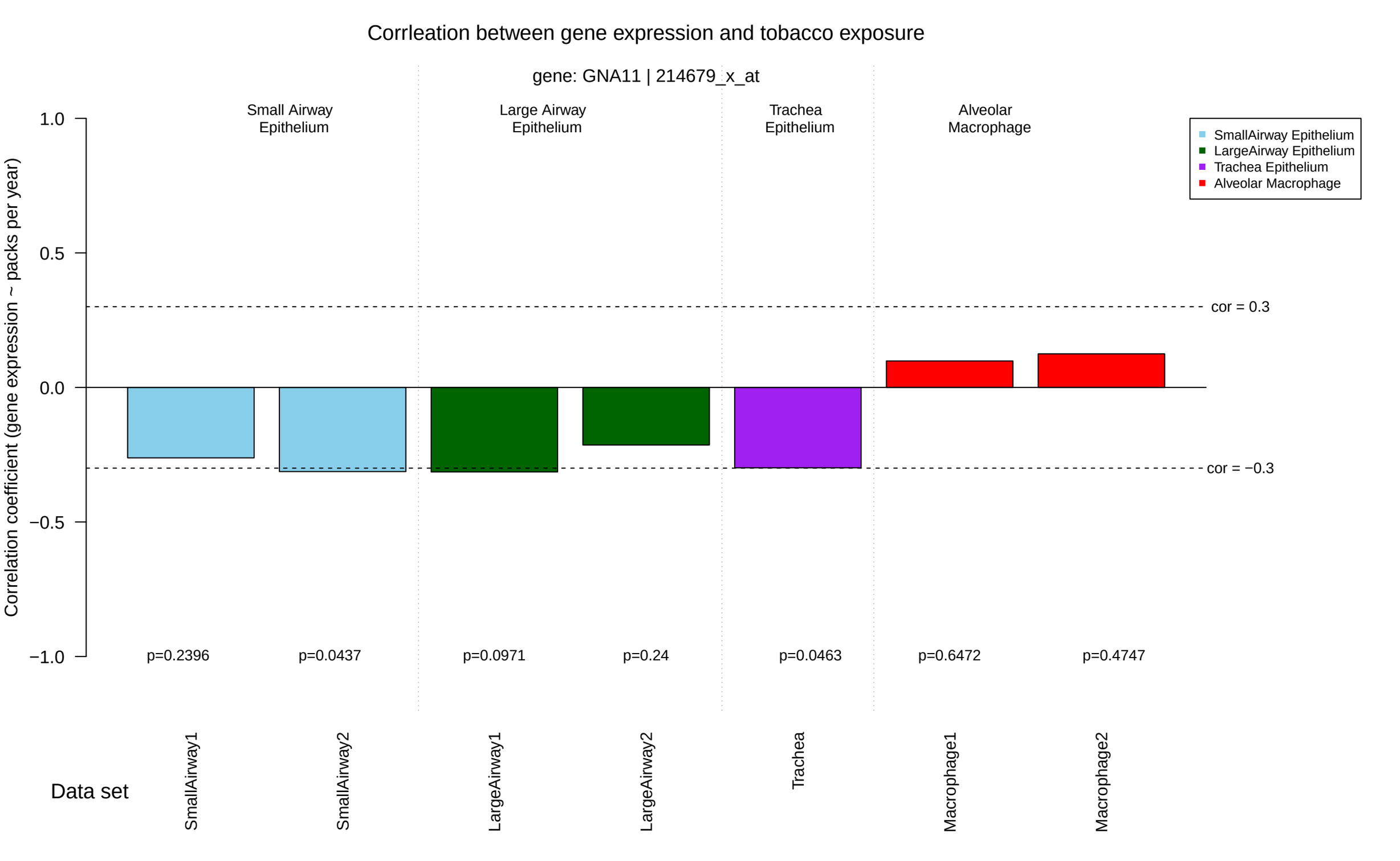

set: LargeAirway1 (p=0.1334)
GNA11 | 214679_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.49)
GNA11 | 214679_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0099)
GNA11 | 214679_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9908)
GNA11 | 214679_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1839)
GNA11 | 214679_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNA11 | 214679_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2396
p=0.0437
p=0.0971
p=0.24
p=0.0463
p=0.6472
p=0.4747
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2